

IN-HOUSE ADHESIVES & EPOXY

Complete Technical Guide

Underfill • Stacking Adhesive • Pressure Sensitive Adhesive (PSA)
RTV Silicones • Encapsulation Materials • Process Standards

PCBA Manufacturing & Assembly Standards
IPC/J-STD-001 | IPC-A-610 | MIL-STD-883 | RoHS

1. IN-HOUSE ADHESIVE & EPOXY OVERVIEW

This guide documents all adhesives, epoxies, and encapsulants currently stocked and used in-house for PCBA manufacturing. These materials serve critical functions including mechanical reinforcement, component stacking, environmental protection, and electrical insulation. Proper selection and application ensure long-term reliability for automotive (Tesla, Zoox), medical, and industrial applications.

In-House Material Categories:

- **Underfill Materials:** Capillary flow epoxies for BGA/CSP reinforcement
- **Stacking Adhesives:** Structural epoxies for component-on-component assembly
- **Pressure Sensitive Adhesives (PSA):** Bonding without heat or solvent activation
- **RTV Silicones:** Room Temperature Vulcanizing encapsulants for environmental protection

Application Methods:

- **Dispensing:** Pneumatic or positive displacement dispensers
- **Stencil Printing:** For controlled adhesive volume on large areas
- **Manual Application:** Syringe or brush for rework/repair
- **Lamination:** For PSA tapes and films

■ **Key Principle:** Adhesive selection depends on five factors: **cure mechanism** (thermal, UV, moisture, RTV), **viscosity** (dispensing vs. flow), **thermal conductivity** (heat dissipation), **electrical properties** (insulation vs. conductivity), and **environmental resistance** (temperature, humidity, chemicals).

2. LOCTITE 3549 — UNDERFILL MATERIAL

2.1 Product Identification

LOCTITE® 3549 is a **one-component, heat-cure epoxy underfill** specifically designed for CSP (Chip Scale Package) and BGA (Ball Grid Array) reinforcement. It flows by capillary action beneath the component and cures to form a rigid mechanical bond between the chip and PCB.

Product Specifications:

- **Type:** One-part, heat-cure epoxy underfill
- **Color:** Black (for visual contrast and UV blocking)
- **Viscosity:** Low — designed for capillary flow under components
- **Cure Schedule:** 150°C for 10–15 minutes (typical)
- **Alternative Cure:** 165°C for 5–8 minutes (faster throughput)
- **Glass Transition Temperature (Tg):** ~120–140°C
- **CTE (Coefficient of Thermal Expansion):** Matched to solder and PCB

Key Features:

- **Capillary flow:** Self-levels under BGA/CSP without voids
- **Mechanical reinforcement:** Reduces solder joint stress from thermal cycling
- **Drop shock improvement:** Increases impact resistance of portable devices
- **Low CTE:** Minimizes thermal mismatch between silicon and PCB
- **Excellent adhesion:** Bonds to solder mask, solder, and component body

Applications:

- BGA underfill for automotive electronics (Tesla, Zoox)
- CSP reinforcement for smartphones and tablets
- Flip-chip underfill for advanced packaging
- High-reliability assemblies subject to thermal cycling

Process Parameters:

- **Storage:** -40°C to -20°C (frozen); 30-day shelf life at 25°C after thaw
- **Thaw Time:** 1–2 hours at room temperature before use
- **Dispense Pattern:** L-shape or I-shape along component edge
- **Flow Time:** 30–120 seconds depending on gap height and viscosity
- **Cure Oven:** Convection or IR oven at 150°C
- **Inspection:** Acoustic Microscopy (C-SAM) for void detection

■ ■ **Process Caution:** Do NOT pre-heat PCB before underfill dispense. Excessive substrate temperature accelerates cure before complete flow, causing voids and incomplete fill. Maintain board temperature < 40°C during dispense and flow.

3. EA 0151 — STACKING ADHESIVE

3.1 Product Identification

LOCTITE® EA 0151 (formerly Hysol® 0151) is a **thixotropic epoxy adhesive** used for **component stacking** applications. It provides structural bonding between stacked components where mechanical fasteners are not feasible.

Product Specifications:

- **Type:** Two-part or one-part heat-cure epoxy (verify in-house variant)
- **Color:** Yellow/Amber (typical for Hysol epoxies)
- **Viscosity:** Thixotropic — stays in place after dispense, no sag
- **Cure Schedule:** 150°C for 20–30 minutes (typical)
- **Glass Transition Temperature (T_g):** ~80–100°C
- **Shear Strength:** ~15–25 MPa (strong structural bond)

Key Features:

- **Thixotropic:** Does not flow after dispense — ideal for vertical surfaces
- **High strength:** Structural bond for component stacking
- **Good gap filling:** Fills irregular surfaces between components
- **Electrical insulation:** Prevents shorting between stacked parts

Applications:

- Stacking memory modules (DDR on DDR)
- Stacking power modules or RF components
- Bonding heat sinks to component packages
- Structural reinforcement of tall components

Process Parameters:

- **Storage:** 10–25°C (refrigeration not required for most variants)
- **Mix Ratio:** Verify if two-part — typically 1:1 by weight or volume
- **Dispense:** Dot or line pattern on lower component before placement
- **Placement:** Apply upper component within open time (typically 30–60 min)
- **Cure:** Convection oven at 150°C for 20–30 minutes

■ **Best Practice:** For stacking applications, ensure the lower component is fully soldered and cleaned before adhesive application. The adhesive cure temperature must not exceed the reflow temperature of existing solder joints.

4. PSA — PRESSURE SENSITIVE ADHESIVE

4.1 Product Identification

PSA (Pressure Sensitive Adhesive) is a special adhesive material used in-house for applications requiring instant bonding without heat, solvent, or UV activation. PSA forms a bond when pressure is applied to the adhesive surface.

Important Note:

The in-house PSA is a **special adhesive requiring a specific Manufacturer Part Number (MPN)**. The exact PSA variant must be confirmed with the MPN before use. Different PSA formulations have vastly different properties (temperature resistance, adhesion strength, substrate compatibility).

General PSA Specifications (Verify with MPN):

- **Type:** Acrylic or silicone-based pressure sensitive adhesive
- **Form:** Tape, film, or liquid coating on carrier
- **Activation:** Pressure only — no heat, UV, or solvent required
- **Tack:** Instant grab upon contact
- **Peel Strength:** Varies by formulation — verify with MPN datasheet
- **Temperature Range:** Typically -40°C to +150°C (verify with MPN)

Common PSA Applications in PCBA:

- **Shielding tape:** EMI/RFI shielding attachment
- **Insulation pads:** Thermal or electrical insulation between components
- **Label/fixture bonding:** Temporary or permanent fixture attachment
- **Flex circuit lamination:** Bonding flex layers in rigid-flex assemblies
- **Gasket attachment:** Sealing gaskets to enclosure or PCB

Process Parameters:

- **Surface Preparation:** Clean, dry, oil-free surface essential
- **Application Pressure:** 0.1–0.5 MPa (15–75 psi) for 5–10 seconds
- **Dwell Time:** Full adhesion develops over 24–72 hours
- **Temperature:** Room temperature application; verify high-temp MPN for automotive

■ **CRITICAL:** Confirm the exact MPN before using PSA. Using the wrong PSA formulation can result in adhesive failure, outgassing in vacuum environments, or incompatibility with conformal coating. Document MPN on work order and retain certificate of analysis.

5. RTV SILICONES — ENCAPSULATION MATERIALS

RTV (Room Temperature Vulcanizing) silicones are used in-house for **encapsulation** of electronic components. They provide environmental protection (moisture, dust, chemicals) while maintaining electrical insulation and thermal stability. RTVs cure at room temperature via moisture in the air (condensation cure) or via platinum catalyst (addition cure).

Clarification: LOCTITE 0151 is NOT an RTV. The in-house RTV materials are:

- **RTV162-White** (Momentive)
- **LOCTITE SI 595** (Henkel)
- **LOCTITE SI 5011** (Henkel) — Part Number: R1510783X

5.1 RTV162-White (Momentive)

RTV162 is a **white, one-part, condensation-cure silicone adhesive sealant**. It is specifically formulated for electronics encapsulation and meets MIL-A-46146B Type I requirements.

Product Specifications:

- **Type:** One-part, moisture-cure RTV silicone
- **Color:** White
- **Viscosity:** Paste — non-sag, suitable for vertical applications
- **Cure Mechanism:** Condensation cure (releases acetic acid or alcohol)
- **Skin-Over Time:** 10–30 minutes at 25°C, 50% RH
- **Full Cure:** 24 hours at 25°C, 50% RH (7 days for maximum properties)
- **Hardness (Shore A):** ~25–35
- **Temperature Range:** -60°C to +200°C (continuous); +250°C (short term)
- **Dielectric Strength:** ~20 kV/mm
- **MIL Spec:** MIL-A-46146B Type I, Class I

Applications:

- Encapsulation of power transistors and ICs
- Potting of connectors and terminal blocks
- Sealing of enclosures and housings
- Thermal insulation of heat-sensitive components
- Military and aerospace electronics (MIL-spec compliance)

Process Parameters:

- **Storage:** 10–25°C, sealed tube; 12-month shelf life
- **Surface Prep:** Clean, dry, grease-free surface
- **Application:** Dispense from tube or cartridge; tool with spatula
- **Cure Acceleration:** Elevated humidity (60–70% RH) speeds cure

- **Cure Inhibition:** Avoid contact with tin, sulfur, or amine-containing materials

5.2 LOCTITE SI 595

LOCTITE® SI 595 is a **clear, one-part, acetoxycure RTV silicone adhesive sealant**. It offers excellent clarity for applications where visual inspection of encapsulated components is required.

Product Specifications:

- **Type:** One-part, moisture-cure RTV silicone
- **Color:** Clear (transparent)
- **Viscosity:** Paste — non-sag
- **Cure Mechanism:** Acetoxycure (releases acetic acid during cure)
- **Skin-Over Time:** 10–25 minutes at 25°C, 50% RH
- **Full Cure:** 24 hours at 25°C, 50% RH
- **Hardness (Shore A):** ~20–30
- **Temperature Range:** -55°C to +200°C
- **Dielectric Strength:** ~18 kV/mm

Applications:

- Clear encapsulation for LED assemblies
- Potting where visual inspection of components is required
- Sealing of transparent enclosures or displays
- General electronics encapsulation with aesthetic requirements

Process Parameters:

- **Storage:** 10–25°C, sealed tube; 12-month shelf life
- **Surface Prep:** Clean, dry; acetic acid release may corrode sensitive metals
- **Application:** Dispense and tool to desired shape
- **Cure Acceleration:** Higher humidity accelerates cure
- ■■ **Caution:** Acetic acid vapor during cure — ensure ventilation

5.3 LOCTITE SI 5011 — Part Number R1510783X

LOCTITE® SI 5011 (Part Number: **R1510783X**) is a **non-corrosive, one-part, neutral-cure RTV silicone adhesive sealant**. It is specifically formulated for sensitive electronics where acetic acid or alcohol release would damage components.

Product Specifications:

- **Type:** One-part, neutral-cure (oxime-cure) RTV silicone
- **Color:** Clear (CL variant) or other colors (verify R1510783X spec)
- **Viscosity:** Paste — non-sag
- **Cure Mechanism:** Neutral cure (releases alcohol — non-corrosive)
- **Skin-Over Time:** 15–30 minutes at 25°C, 50% RH
- **Full Cure:** 24 hours at 25°C, 50% RH
- **Hardness (Shore A):** ~25–35
- **Temperature Range:** -55°C to +200°C
- **Dielectric Strength:** ~20 kV/mm
- **Key Feature:** Non-corrosive to metals (copper, brass, zinc)

Applications:

- Encapsulation of sensitive copper or silver components
- Potting near connectors with mixed metal contacts
- Sealing of enclosures with corrosion-sensitive hardware
- General electronics where non-corrosive cure is mandatory

Process Parameters:

- **Storage:** 10–25°C, sealed tube/cartridge; 12-month shelf life
- **Surface Prep:** Clean, dry, grease-free
- **Application:** Dispense from 300 mL cartridge or tube
- **Cure Acceleration:** 60–70% RH optimal; heat to 60–80°C for faster cure
- **Compatibility:** Safe on copper, aluminum, steel, most plastics

■ **In-House Recommendation:** Use SI 5011 (R1510783X) as the default RTV for general encapsulation. Use RTV162-White for military/aerospace applications requiring MIL-spec compliance. Use SI 595 only when optical clarity is required (note acetic acid caution).

6. RTV SILICONE COMPARISON TABLE

Property	RTV162-White	LOCTITE SI 595	LOCTITE SI 5011 (R1510783X)
Manufacturer	Momentive	Henkel	Henkel
Color	White	Clear	Clear
Cure Type	Condensation (alcohol)	Acetoxy (acetic acid)	Neutral (oxime)
Corrosivity	Low	Moderate (acetic acid)	Non-corrosive
Skin-Over Time	10–30 min	10–25 min	15–30 min
Full Cure	24 hr / 7 days max	24 hr	24 hr
Hardness (Shore A)	25–35	20–30	25–35
Temp Range	-60°C to +200°C	-55°C to +200°C	-55°C to +200°C
Dielectric Strength	~20 kV/mm	~18 kV/mm	~20 kV/mm
MIL Spec	MIL-A-46146B Type I	N/A	N/A
Best For	Military/aerospace Opaque encapsulation	Clear encapsulation LED/visual apps	General electronics Non-corrosive cure

Table 1: In-House RTV Silicone Encapsulants Comparison

7. IN-HOUSE ADHESIVE & EPOXY COMPLETE MATRIX

Material	Type	Category	Cure	Color	Primary Use
LOCTITE 3549	Epoxy	Underfill	Heat 150°C	Black	BGA/CSP reinforcement
EA 0151	Epoxy	Stacking	Heat 150°C	Yellow	Component stacking
PSA	Acrylic/Silicone	PSA	Pressure	Various	Shielding/insulation (MPN required)
RTV162-White	Silicone	Encapsulation	RTV Moisture	White	MIL-spec encapsulation
LOCTITE SI 595	Silicone	Encapsulation	RTV Acetoxy	Clear	Clear encapsulation
LOCTITE SI 5011 (R1510783X)	Silicone	Encapsulation	RTV Neutral	Clear	General non-corrosive encapsulation

Table 2: In-House Adhesive & Epoxy Complete Matrix

8. STORAGE, HANDLING & SAFETY

8.1 Storage Requirements

LOCTITE 3549 (Underfill):

- **Temperature:** -40°C to -20°C (frozen storage)
- **Shelf Life:** 6 months frozen; 30 days at 25°C after thaw
- **Thaw:** 1–2 hours at room temperature before use
- **Do NOT refreeze:** Once thawed, use within open time or discard

EA 0151 (Stacking):

- **Temperature:** 10–25°C (ambient, dry)
- **Shelf Life:** 12 months (verify TDS)
- **Seal:** Keep container tightly closed when not in use

PSA:

- **Temperature:** 15–25°C, away from direct sunlight
- **Shelf Life:** Varies by MPN — verify datasheet
- **Protection:** Keep release liner intact until application

RTV Silicones (RTV162, SI 595, SI 5011):

- **Temperature:** 10–25°C, sealed tube/cartridge
- **Shelf Life:** 12 months (unopened)
- **Humidity:** Store in dry environment — moisture initiates cure
- **Opened tubes:** Use within 30 days; cure from nozzle inward

8.2 Safety Precautions

- **Skin contact:** Wear nitrile gloves when handling all adhesives
- **Eye protection:** Safety glasses mandatory during dispense
- **Ventilation:** Ensure adequate airflow — especially for acetoxo RTVs
- **LOCTITE 3549:** Contains epoxy — sensitizer; avoid repeated skin contact
- **RTV162:** Alcohol release during cure — flammable vapors
- **SI 595:** Acetic acid release — corrosive vapor, ensure exhaust
- **SI 5011:** Oxime release — use in ventilated area

■ **CRITICAL SAFETY:** Epoxy materials (3549, EA 0151) can cause allergic sensitization with repeated skin contact. Always wear gloves and wash hands after handling. If skin contact occurs, wash immediately with soap and water. Seek medical attention if irritation persists.

9. APPLICATION PROCESS GUIDELINES

9.1 Underfill (LOCTITE 3549) Process

1. **Solder:** Complete reflow soldering of BGA/CSP first
2. **Clean:** Clean board if water-soluble flux used (optional for no-clean)
3. **Preheat:** Warm board to 40–60°C (reduces viscosity, improves flow)
4. **Dispense:** Apply along one or two edges of component
5. **Flow:** Allow 30–120 seconds for capillary fill under component
6. **Inspect:** Check fillet formation around component edges
7. **Cure:** 150°C for 10–15 minutes in convection oven
8. **Verify:** Acoustic microscopy (C-SAM) for void detection (Class 3)

9.2 Stacking Adhesive (EA 0151) Process

1. **Prepare:** Ensure lower component is soldered and cleaned
2. **Dispense:** Apply dots or lines on lower component body
3. **Place:** Align upper component within 30–60 minutes (open time)
4. **Press:** Apply gentle pressure to spread adhesive
5. **Cure:** 150°C for 20–30 minutes in convection oven
6. **Inspect:** Verify no adhesive overflow onto solder pads

9.3 PSA Application Process

1. **Verify MPN:** Confirm correct PSA material per work order
2. **Cut:** Cut PSA to required shape and size
3. **Peel:** Remove release liner from one side
4. **Position:** Align PSA on substrate or component
5. **Apply Pressure:** 0.1–0.5 MPa for 5–10 seconds
6. **Remove Second Liner:** If double-sided, peel second liner
7. **Bond:** Apply second substrate and press again
8. **Dwell:** Allow 24–72 hours for full adhesion development

9.4 RTV Encapsulation Process

1. **Prepare:** Clean and dry target area; mask areas not to be coated
2. **Dispense:** Apply RTV from tube, cartridge, or automated dispenser
3. **Tool:** Shape with spatula or automated profiling
4. **Level:** Ensure complete coverage of target component/area
5. **Cure:** 24 hours at 25°C, 50% RH (or accelerated at 60–80°C)
6. **Inspect:** Verify no voids, bubbles, or uncovered areas
7. **Test:** Electrical test after full cure (7 days for maximum properties)

■ **In-House Best Practice:** For automotive production (Tesla, Zoox), always verify adhesive cure completeness with DSC (Differential Scanning Calorimetry) or DMA (Dynamic Mechanical Analysis) on first article. Document cure profile and adhesive lot number per work order for traceability.

10. QUICK REFERENCE SUMMARY

Underfill:

- **LOCTITE 3549** — Black epoxy, 150°C cure, BGA/CSP reinforcement
- **Storage:** Frozen -40°C; thaw 1–2 hr; use within 30 days
- **Flow:** Capillary underfill; cure 150°C/10–15 min

Stacking Adhesive:

- **EA 0151** — Yellow thixotropic epoxy, 150°C cure
- **Storage:** Ambient 10–25°C; 12-month shelf life
- **Application:** Dot/line dispense; place upper component; cure 150°C/20–30 min

Pressure Sensitive Adhesive:

- **PSA** — MPN REQUIRED before use
- **Activation:** Pressure only (0.1–0.5 MPa)
- **Dwell:** 24–72 hours for full adhesion

RTV Encapsulants:

- **RTV162-White** — White, MIL-A-46146B, opaque encapsulation
- **LOCTITE SI 595** — Clear, acetoxycure, visual applications
- **LOCTITE SI 5011 (R1510783X)** — Clear, neutral cure, non-corrosive, general use
- **Storage:** 10–25°C sealed; 12 months; moisture initiates cure
- **Cure:** 24 hr at 25°C/50% RH; 7 days for max properties

Safety Reminders:

- **Gloves:** Nitrile for all adhesive handling
- **Ventilation:** Mandatory for acetoxycure and alcohol-cure RTVs
- **Eye protection:** Safety glasses during dispense
- **Epoxy sensitization:** Avoid repeated skin contact with 3549 and EA 0151

References & Standards:

- IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- IPC-A-610: Acceptability of Electronic Assemblies
- MIL-A-46146B: Adhesive, Silicone, Room Temperature Vulcanizing (RTV)
- MIL-STD-883: Test Methods and Procedures for Microelectronics
- Henkel LOCTITE Technical Datasheets: 3549, EA 0151, SI 595, SI 5011
- Momentive RTV162 Technical Datasheet
- PSA Manufacturer Datasheet (MPN Required)

End of Document — In-House Adhesives & Epoxy Complete Guide